

CCNA 200-301: Class Notes — Day 9

Topic: Inside a Real Data Center & Hands-On with Cisco Equipment

Modern networking requires both theoretical facility knowledge and hands-on operational capability. Network engineers must understand data center physical infrastructure, thermal/power architecture, copper and optical fiber cabling standards, physical termination tools, and console management software for hardware configuration.

1. Data Center Tier Classifications (Uptime Institute)

Data centers are classified into four distinct Tiers based on operational uptime, redundancy, and fault tolerance:

Tier Level	Expected Uptime	Annual Downtime	Redundancy Level	Key Characteristic
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Tier I	99.671%	< 28.8 hours	N (No redundancy)	Single path for power and cooling; prone to disruption from planned or unplanned events.
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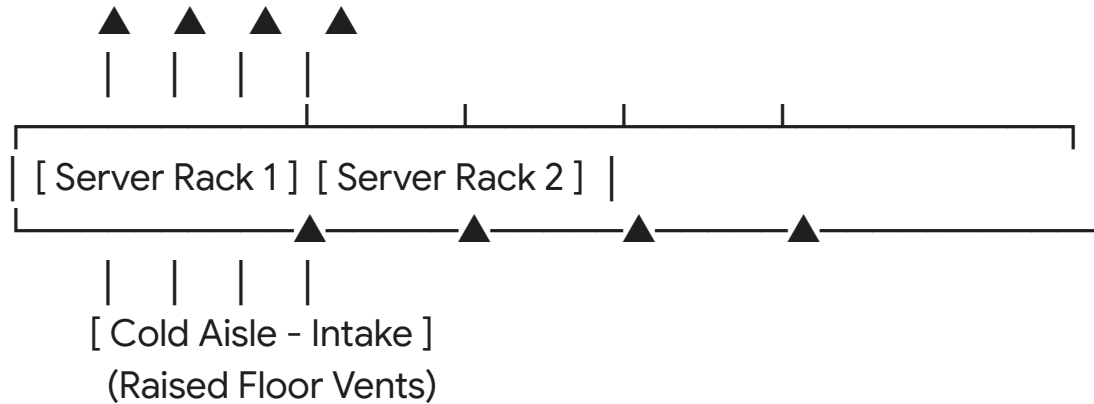
Tier II	99.741%	< 22.0 hours	$N + 1$ (Component redundancy)	Redundant power and cooling components, but single distribution path.
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Tier III	99.982%	< 1.6 hours	$N + 1$ (Concurrently Maintainable)	Multiple power and cooling paths; hardware can be serviced without shutting down IT operations.
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Tier IV	99.995%	< 26.3 minutes	$2N + 1$ (Fault Tolerant)	Fully compartmentalized, dual-powered independent paths; tolerates single worst-case failure without impact.
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2. Physical Layout & Thermal Management

[Hot Aisle - Exhaust]

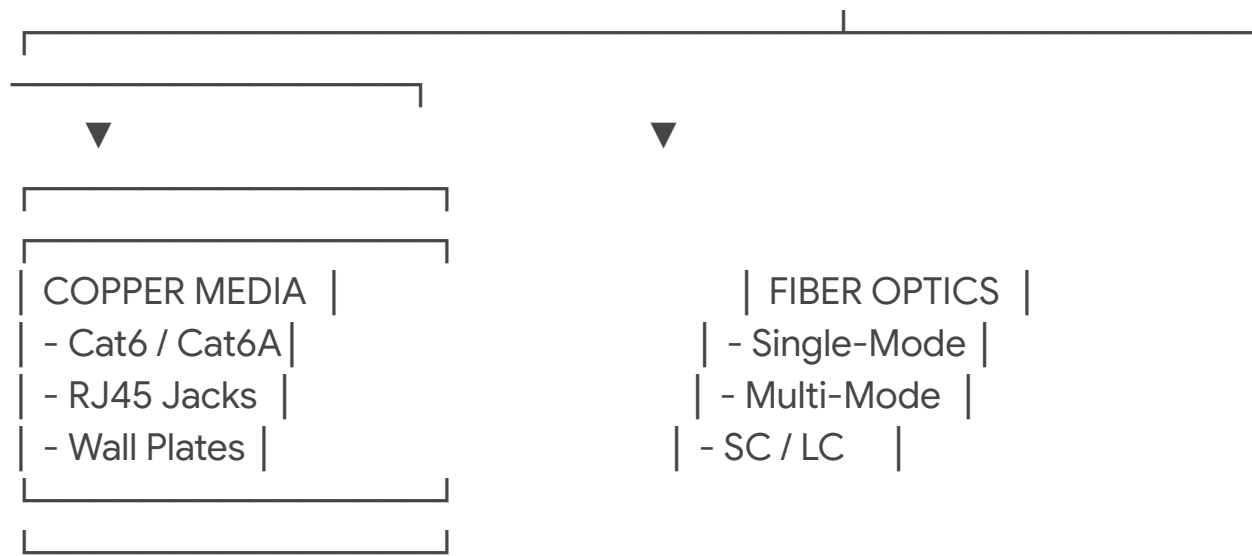


- **Rack Unit (RU / U):** Standard vertical measurement unit for mounting equipment ($1\text{ U} = 1.75\text{ inches}$ or 44.45 mm). Standard enterprise racks are 19 inches wide and 42U or 48U tall.
- **Hot Aisle / Cold Aisle Topology:** Arranging server racks in alternating rows (intake facing intake, exhaust facing exhaust) combined with Cold Aisle Containment (CAC) or Hot Aisle Containment (HAC) to optimize HVAC efficiency according to **ASHRAE standards** ($18^{\circ}\text{C} - 27^{\circ}\text{C}$).
- **Power Redundancy Chain:** Utility Grid $\rightarrow$ Automatic Transfer Switch (ATS) $\rightarrow$ Uninterruptible Power Supply (UPS) $\rightarrow$ Diesel Generator $\rightarrow$ Rack Power Distribution Unit (PDU).

3. Physical Cabling & Connector Breakdown

Connecting networking hardware across structured enterprise environments requires understanding copper Ethernet standards, optical fiber types, and physical connectors.





A. Copper Ethernet Cabling & RJ45

- **UTP / STP Cable (Cat5e / Cat6 / Cat6A):** Unshielded/Shielded Twisted Pair copper wire designed for balanced signal transmission up to 100 meters (328 feet).
- **RJ45 Port & Connector:** Registered Jack 45 (8P8C modular connector) used to terminate copper Ethernet cables for switches, routers, and host PCs.
- **Wall Plate / Keystone Jack:** Modular wall-mounted or patch-panel outlet housing RJ45 keystone sockets where horizontal cabling drops terminate inside office work areas.
- **PSE & PoE Cable (Power Sourcing Equipment):** Switches or injectors supplying Direct Current (DC) power alongside network data over copper Cat cables to Powered Devices (PD) such as IP Phones, Access Points, and IP Cameras (IEEE 802.3af / 802.3at / 802.3bt).

B. Fiber Optic Cabling & Optical Connectors

Optical fiber transmits data as light pulses through glass strands, offering immunity to Electromagnetic Interference (EMI) and extreme

long-distance range.

| **Connector / Cable Type** | **Full Name / Descriptor** | **Physical Characteristics & Typical Use Case** || **Single-Mode Fiber (SMF)** | Yellow Outer Jacket ($9/125\ \mu\text{m}$) | Uses laser light over a tiny core for long-haul WAN/campus runs ($> 10\ \text{km}$). || **Multi-Mode Fiber (MMF)** | Aqua/Orange Jacket ($50/125\ \mu\text{m}$) | Uses LED/VCSEL light over a larger core for short-range intra-datacenter links ($< 550\ \text{meters}$). || **LC Connector** | **Lucent Connector** ("Little Connector") | Small form factor push-pull connector; standard for high-density SFP/SFP+ transceiver modules on switches and routers. || **SC Connector** | **Subscriber Connector** ("Square Connector") | Snap-in push-pull square connector commonly used in FTTH CPE drops, media converters, and legacy patch panels. |

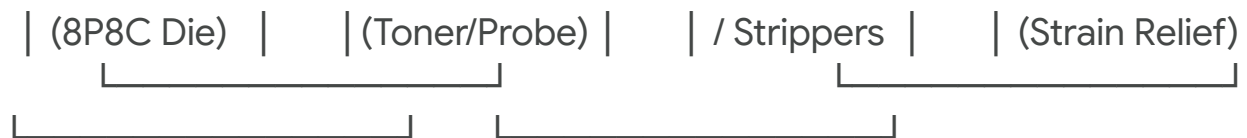
C. Optical Conversion & Termination Hardware

- **Media Converter:** Transceiver unit that converts electrical signals from a copper RJ45 Cat cable into optical signals for long-distance fiber transmission (and vice versa).
- **Fiber Termination Box (FTB) / ODF:** A wall-mounted or rack-mounted enclosure used to protect, route, and organize optical fiber splices, pigtails, and coupler panels.
- **Cable Manager:** Horizontal and vertical rack accessories (D-rings, brush panels, finger ducts) that hold copper patch cords and fiber jumpers neatly to ensure proper bend radius and unobstructed airflow.

4. Field Toolkit for Network Technicians

Network engineers and field technicians rely on practical tools to fabricate, test, and trace physical cable drops during infrastructure rollouts:

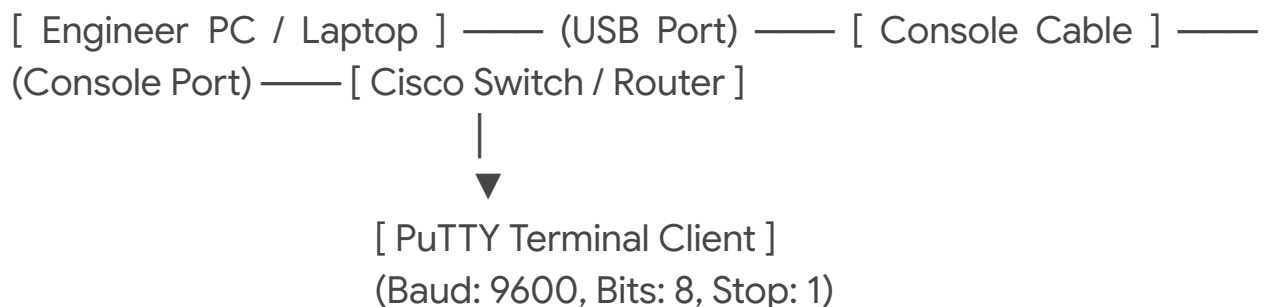




1. **Crimping Tool (Crimper):** Heavy-duty hand tool with an 8P8C die used to compress metal contact pins into copper wire strands inside an RJ45 modular plug (following T568A or T568B pinout standards).
2. **Wire Tracker (Toner & Probe / Cable Tester):**
 - *Tone Generator:* Sends an electrical audio signal down a specific wire pair.
 - *Inductive Probe:* Wirelessly detects the audio tone to locate and trace specific cables through bundled conduits or wall spaces.
 - *Continuity Tester:* Verifies correct pin mapping, split pairs, open circuits, or shorts.
3. **Wire Stripper & Cable Snips ("Snips/Snag"):**
 - *Wire Stripper:* Precision blades that score outer PVC cable jackets without damaging internal twisted pairs.
 - *Snagless Boot:* Rubber protective sleeve fitted over RJ45 plugs to prevent retaining clips from catching or snapping during cable pulls.

5. Cisco Hardware & Out-of-Band Terminal Management

Configuring enterprise switches and routers requires establishing Out-of-Band (OOB) administrative connectivity.



A. Core Networking Hardware

- **Cisco Enterprise Switch:** Layer 2/3 interconnect device connecting endpoints, servers, and VLAN drops within a campus LAN (e.g., Catalyst 9200/9300).
- **Cisco Enterprise Router:** Layer 3 device providing WAN edge connectivity, inter-VLAN routing, NAT, and dynamic protocol routing (e.g., ISR 4000 / Catalyst 8300).

B. Physical Console Cable Connections

- **Rollover Console Cable:** DB9-to-RJ45 cable or modern USB-to-RJ45 cable (equipped with an integrated FTDI serial chip) that bridges a technician's PC directly to the device's blue **CONSOLE** port.
- **Purpose:** Allows CLI access when the network interface has no IP address configured or when the network interface is down.

C. PuTTY Terminal Emulation Setup

PuTTY is an open-source serial and network terminal emulator used to access Cisco IOS CLI.

- **Serial Connection Parameters (Default Cisco Standard):**
 - **Serial Line:** COM1 / COM3 (assigned by Device Manager)
 - **Speed (Baud Rate):** 9600
 - **Data Bits:** 8
 - **Parity:** None
 - **Stop Bits:** 1
 - **Flow Control:** None

6. Cisco Packet Tracer Lab Emulation

For simulation and pre-deployment testing, **Cisco Packet Tracer** provides a virtual sandbox for modeling networks.

[Router0 (ISR 4331)] ===== (GigabitEthernet 0/0/0) ===== [Switch0 (2960)]



[PC-A] [PC-B]

Initial Cisco IOS Command CLI Hierarchy

```
Router> enable                !-- Navigate to Privileged EXEC Mode
Router# configure terminal      !-- Navigate to Global Configuration
Mode
Router(config)# hostname R1     !-- Assign Device Hostname
R1(config)# interface g0/0/0    !-- Enter Interface Sub-configuration
Mode
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown      !-- Enable Physical Interface
R1(config-if)# exit
R1(config)# line console 0      !-- Secure Console Port
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# end
R1# copy running-config startup-config !-- Save Configuration to NVRAM
```

Summary Tool & Equipment Reference Matrix

| **Equipment / Tool** | **Category** | **Primary Network Function** | | **Console Cable** | Out-of-Band Management | Connects PC serial/USB port to Cisco device Console port. | | **PuTTY Software** | Terminal Emulator | Software GUI used to access Cisco CLI via Serial (9600 baud) or SSH. | | **Crimper** | Termination Tool | Attaches RJ45 plugs to copper Cat6 twisted-pair cable ends. | | **Wire Tracker** | Diagnostic Tool | Traces individual cable runs and tests pin continuity. | | **Media Converter** | Signal Conversion | Bridges copper Ethernet (RJ45) and fiber optic (SC/LC) segments. | | **Fiber Termination Box** | Fiber Infrastructure | Encloses optical splices and provides coupler patch points. | | **PSE Cat Cable** | Power Delivery | Supplies PoE power over Cat6 cable to APs and IP Phones. | | **Cisco Packet Tracer** | Lab Simulator | Visual simulation platform for building topologies and

testing CLI commands. |